

Submitted by
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Lab Assignment #07

Aim:

Design a microstrip line and analyze its behavior with matched and mismatched load on CST

Theory:

The microstrip line is a type of electrical transmission line used to convey radio-frequency (RF) signals, consisting of a conducting strip separated from a ground plane by a dielectric substrate. The fundamental behavior of this structure is governed by its characteristic impedance (Z_0), which is determined by the physical geometry—specifically the width of the strip (W), the thickness of the substrate (h), and the relative permittivity (ϵ_r) of the material. Unlike coaxial cables, microstrip lines do not support a pure Transverse Electromagnetic (TEM) mode because the electromagnetic fields exist both within the dielectric and in the surrounding air; instead, they support a quasi-TEM mode. This leads to the concept of effective permittivity (ϵ_{eff}), a weighted average that accounts for the field distribution in both media, directly influencing the phase velocity and wavelength of the signal as it propagates.

In a practical RF system, signal integrity depends heavily on impedance matching. When a microstrip line is terminated with a matched load (where the load impedance Z_L equals the characteristic impedance Z_0), the maximum power is transferred to the load with zero reflection. This state is characterized by a Scattering Parameter (S_{11}) approaching negative infinity in decibels and a Voltage Standing Wave Ratio (VSWR) of 1.0. Under these conditions, the electromagnetic wave travels unidirectionally toward the load, resulting in a uniform electric field distribution along the length of the line and minimal energy leakage, as evidenced by low farfield directivity and radiation efficiency.

Conversely, a mismatched load ($Z_L \neq Z_0$) causes a portion of the incident wave to be reflected back toward the source. This reflection leads to the formation of standing waves, which are stationary patterns of voltage and current along the line created by the constructive and destructive interference of the incident and reflected waves. The severity of this mismatch is quantified by the reflection coefficient (Γ), which is plotted on a Smith Chart to visualize how far the load impedance deviates from the normalized center point. High VSWR and elevated S_{11} values in a mismatched system indicate significant power loss and potential damage to source components, making the study of these parameters essential for designing stable microwave circuits and antennas.

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The following table summarizes the final dimensions applied to the 3D electromagnetic simulation model to achieve precise resonance and impedance matching at 2.4 GHz.

Table 1 - Parametric Details of 3D Model

Parameter Description	Final Value
Operating Frequency	2.4 GHz
Dielectric Constant (Alumina)	9.9
Target Characteristic Impedance	60 Ohm
Substrate Width	30.0 mm
Substrate Length	30.0 mm
Substrate Thickness	1.0 mm
Microstrip Trace Width	0.6364 mm
Microstrip Trace Length	30.0 mm
Conductor Material	PEC
Ground Plane Material	PEC
Conductor Thickness	0.035 mm
Operating Frequency	2.4 GHz
Dielectric Constant (Alumina)	9.9

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Procedure:

1. Open CST Studio Suite and select the New Template option.
2. Select Microwaves & RF / Optical, then Circuits & Components, and choose Transmission Lines.
3. Select the Time Domain Solver to allow for broadband frequency analysis.
4. Set the project units to mm for dimensions and GHz for frequency.
5. Define the simulation frequency range from 1 GHz to 5 GHz to observe the 2.4 GHz center frequency.
6. Use the Brick tool to create an Alumina substrate ($\epsilon_r=9.9$) with dimensions of 30×30×1 mm.
7. Create a ground plane on the bottom face of the substrate using a PEC Brick with a thickness of 0.035 mm.
8. Construct the microstrip trace on the top face of the substrate with a width of 0.6362 mm for a 60 Ω characteristic impedance.
9. Use the Pick Face tool to select the front cross-section of the microstrip trace.
10. Navigate to Macros > Solver > Ports > Calculate Port Extension Coefficient to define the input Waveguide Port.
11. Use the Pick Point tool to select the end of the trace and the corresponding point on the ground plane.
12. Define a Discrete Port between these points, setting the type to S-Parameter and the impedance to 60 Ω for the matched reference run.
13. Run the Time Domain Solver and record the matched S11 and VSWR values.
14. Modify the Discrete Port settings to simulate the mismatched condition by changing the type to Lumped Element.
15. Select the Inductance (L) property and enter 8 mH as the load value.
16. Define Field Monitors for the E-field and H-field at 2.4 GHz to capture the standing wave behavior.
17. Re-run the Time Domain Solver with the inductive load.
18. Observe the S11 plot, noting that it should now be much closer to 0 dB (near-total reflection).
19. Check the Smith Chart to see the impedance point shifted to the far right/top edge, representing a high-inductive mismatch.
20. Visualize the E-field distribution to identify the standing wave pattern (voltage maximums and minimums) along the line.
21. Compare the VSWR of the 8 mH load against the matched 60 Ω load to quantify the magnitude of the mismatch.

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3D Model:

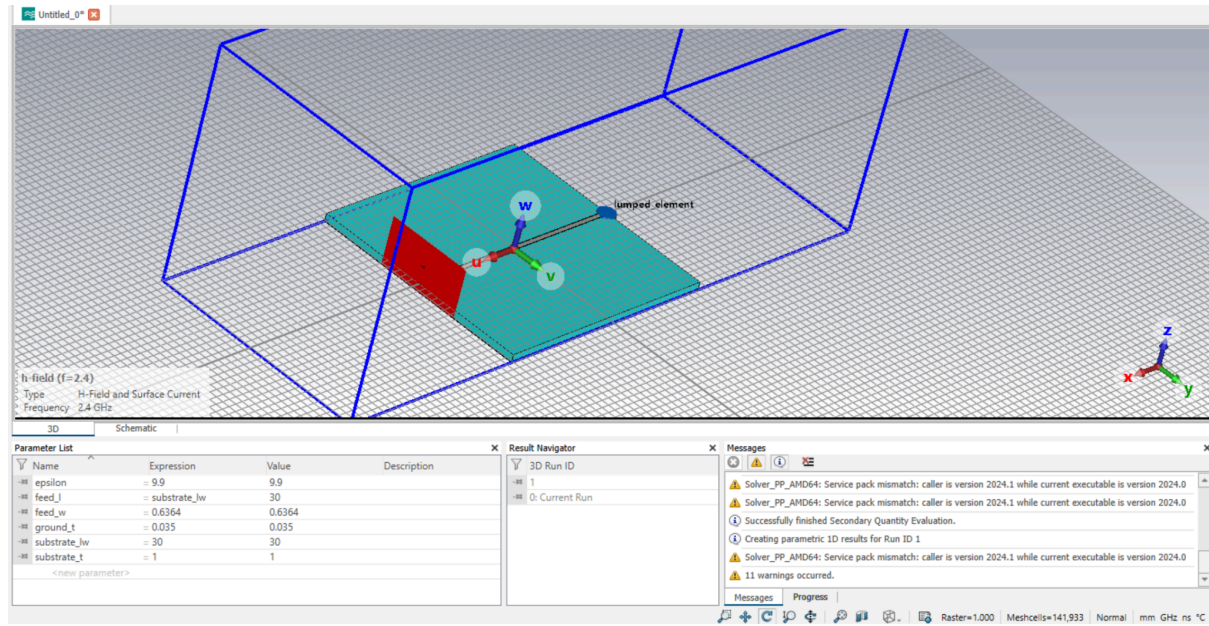


Fig 2.1 Micro Stripline 3D Model with parameters

Results:

Scattering Parameters

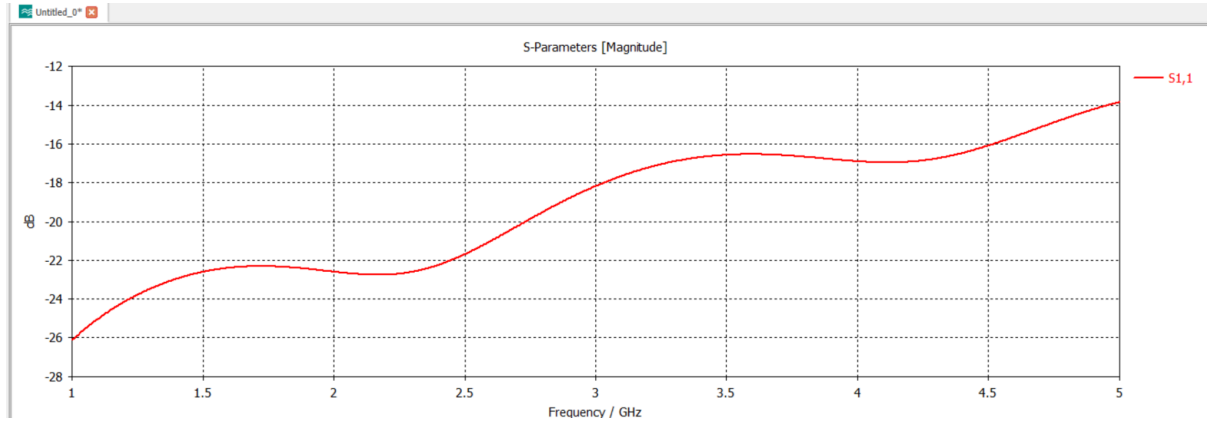


Fig 3.1.1 S parameters of micro stripline (Matched Load 60 Ohm)

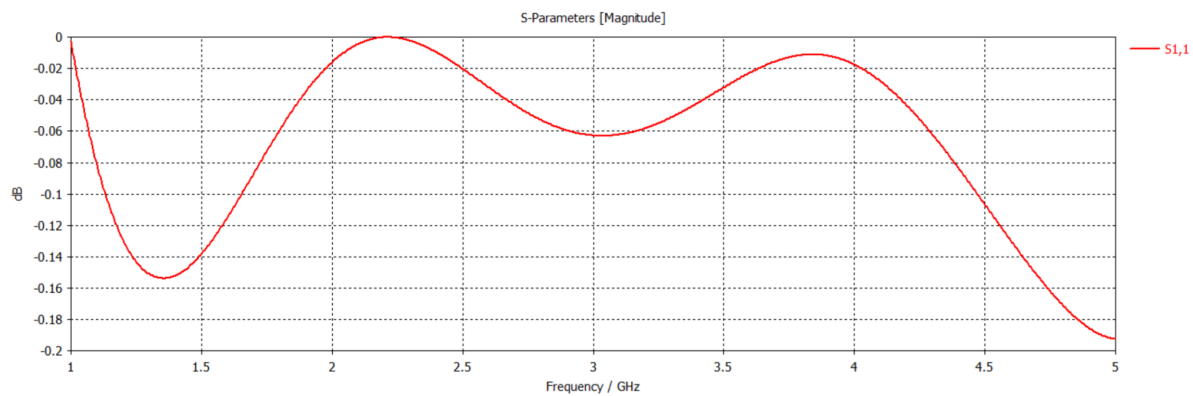


Fig 3.1.2 S parameters of micro stripline (Mismatched Load 8mH)

Reference Impedance

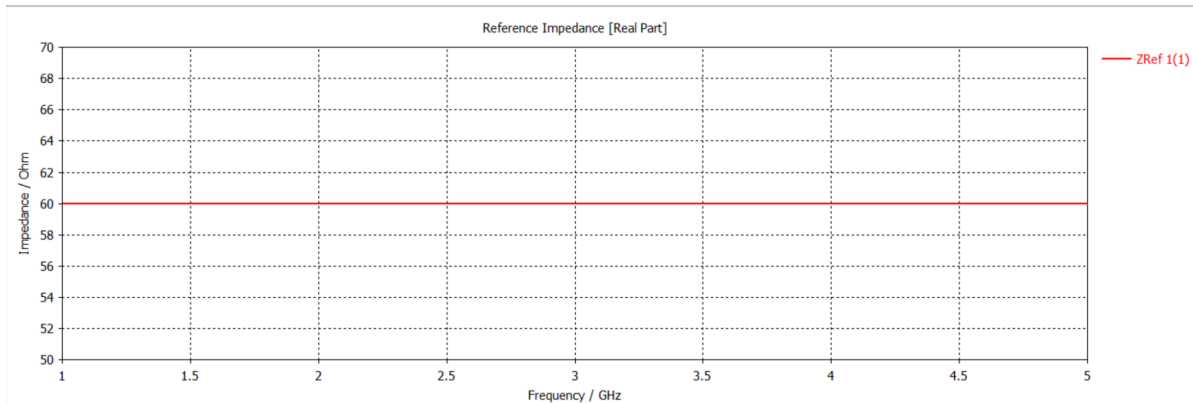


Fig 3.2 Reference impedance

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Voltage Standing Wave Ratio (VSWR)

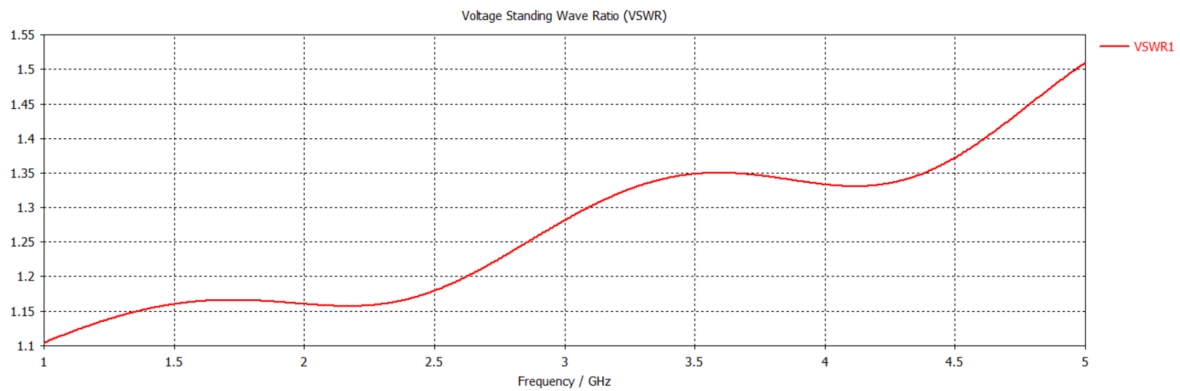


Fig 3.3.1 VSWR of micro stripline (Matched Load 60 Ohm)

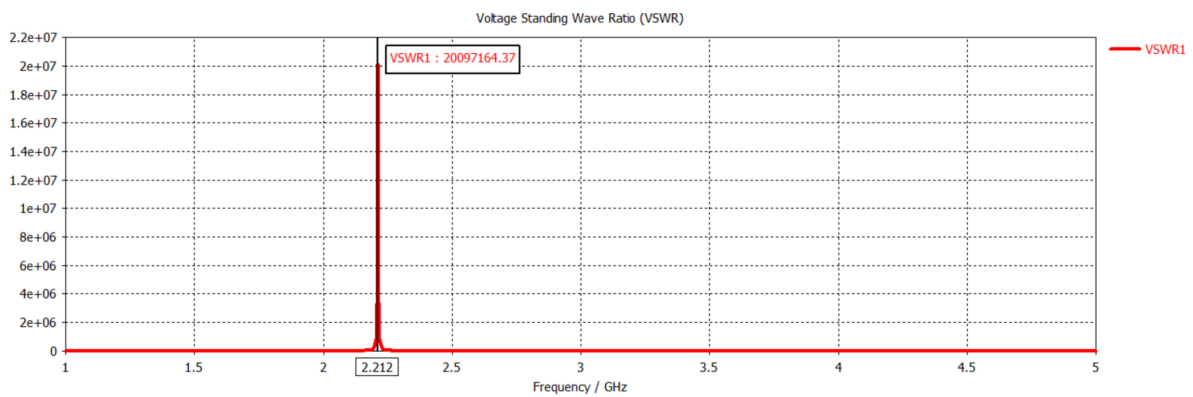


Fig 3.3.2 VSWR of micro stripline (Mismatched Load 8mH)

Farfields

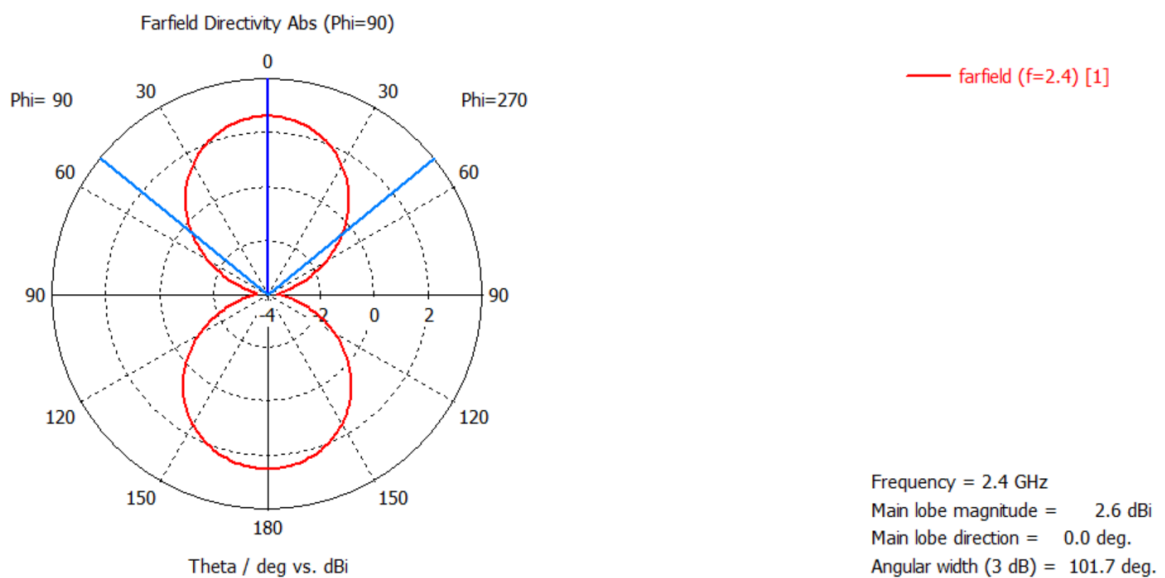


Fig 3.4.1 1D Farfield distribution at 2.4 GHz frequency (Matched Load 60 Ohm)

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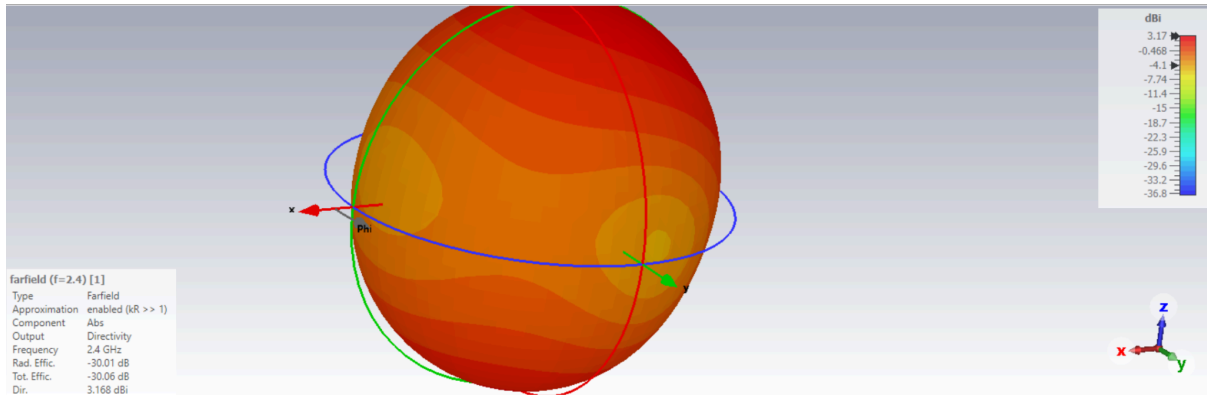


Fig 3.4.2 3D Farfield distribution at 2.4 GHz frequency (Matched Load 60 Ohm)

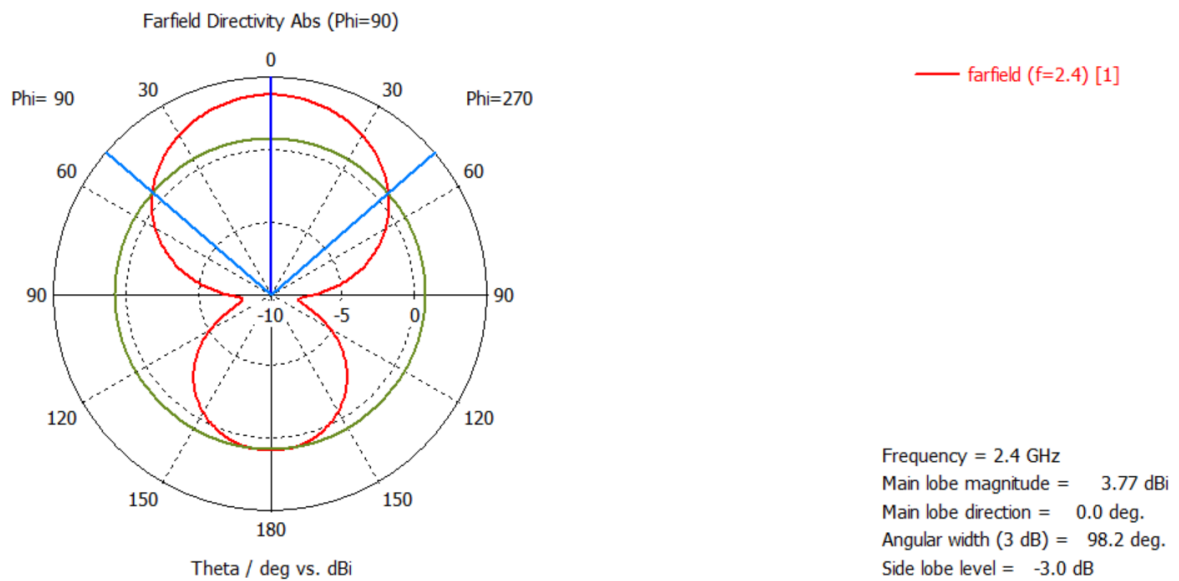


Fig 3.4.3 1D Farfield distribution at 2.4 GHz frequency (Mismatched Load 8mH)

Electric Field Distribution

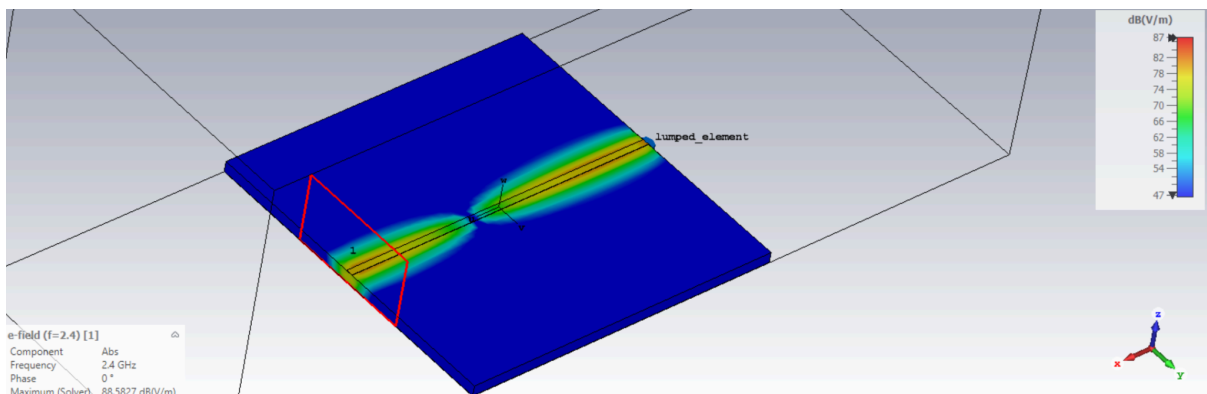


Fig 3.5.1 Electric field distribution at 2.4 GHz frequency (Matched Load 60 Ohm)

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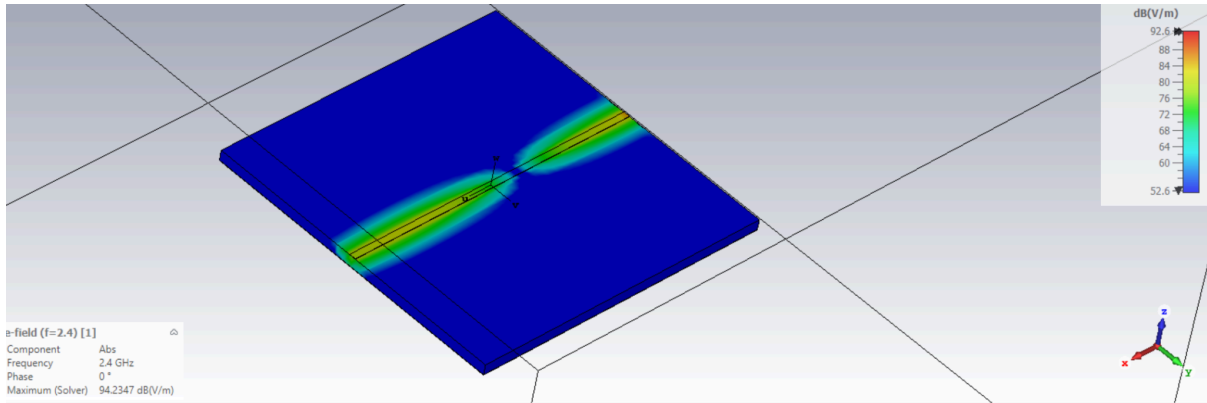


Fig 3.5.2 Electric field distribution at 2.4 GHz frequency (Mismatched Load 8mH)

Magnetic Field Distribution

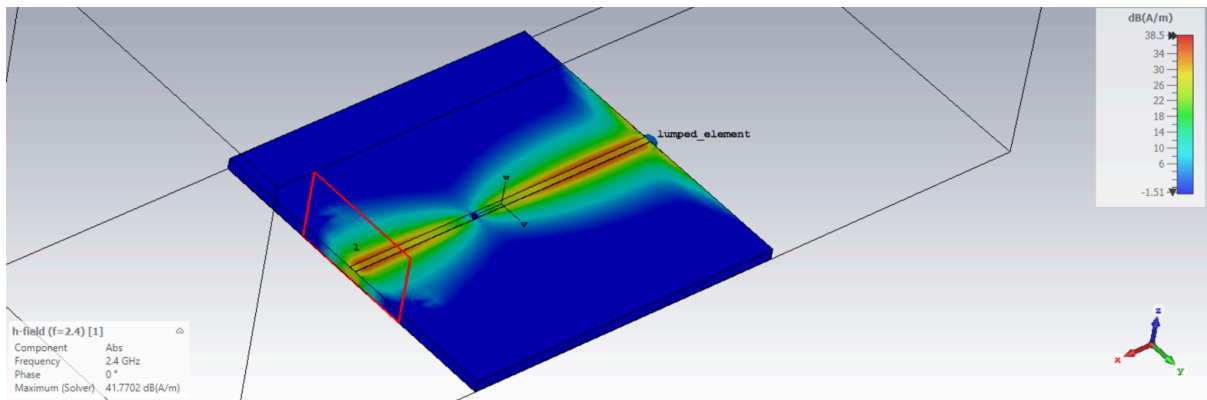


Fig 3.6.1 Magnetic field distribution at 2.4 GHz frequency (Matched Load 60 Ohm)

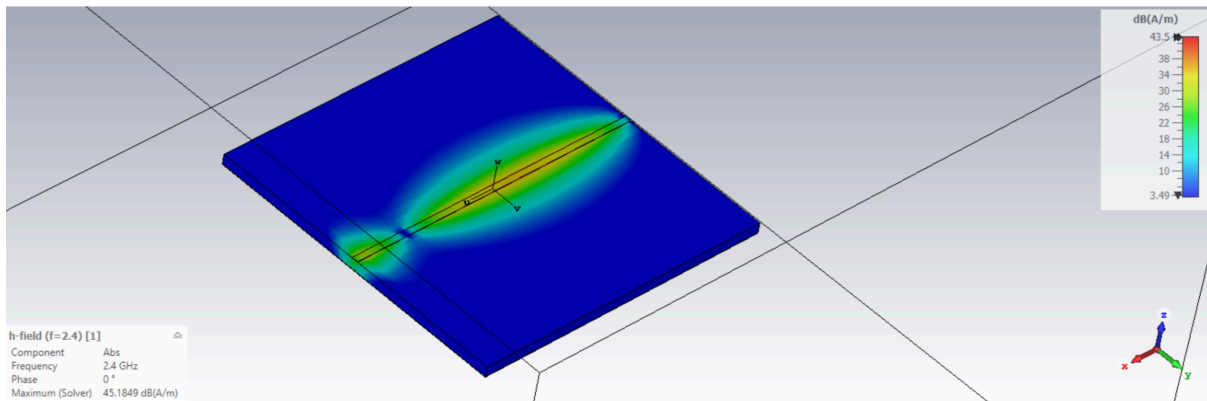


Fig 3.6.2 Magnetic field distribution at 2.4 GHz frequency (Mismatched Load 8mH)

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Conclusions:

In this experiment, a $60\ \Omega$ microstrip transmission line was successfully designed and simulated at a target frequency of 2.4 GHz using an Alumina substrate in CST Studio Suite. The comparative analysis between the matched and mismatched termination states effectively validated fundamental RF transmission line theory. When terminated with a matched $60\ \Omega$ lumped element, the system demonstrated optimal power transfer, characterized by a low return loss (S_{11}) and a Voltage Standing Wave Ratio (VSWR) closely approaching the ideal value of 1.0. Under these matched conditions, the electromagnetic field monitors displayed a uniform, unidirectional traveling wave with negligible radiation leakage. Conversely, terminating the line with a highly inductive 8 mH mismatched load resulted in near-total signal reflection. This severe impedance mismatch was evidenced by the S_{11} magnitude shifting close to 0 dB and a massive resonant spike in the VSWR profile. Furthermore, the electric and magnetic field distributions clearly visualized the formation of standing waves, confirming the presence of stationary voltage and current patterns caused by the interfering incident and reflected waves. Ultimately, this simulation practically demonstrates that precise impedance matching is critical for maintaining signal integrity, maximizing efficiency, and preventing reflected power in high-frequency microwave circuits.